TSAR Project Part 1 – Exploratory Data Analysis

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```r
# # Random sample your group's data set from the original LCdata data set
# LCdata <- fread("LCdata.csv")
# set.seed(1)
# myLCdata <- LCdata %>% sample(5) %>% slice_sample(prop = .5)
# fwrite(myLCdata, file = "myLCdata.csv")
myLCdata <- fread("myLCdata.csv")
```

1 Task 1: Business Understanding

```r
glimpse(myLCdata)
```

```
Rows: 44,369
Columns: 5
$ annual_inc     <dbl> 60000, 18576, 40000, 35000, 125000, 144000, 205000, 118~
$ open_acc       <int> 6, 7, 15, 3, 12, 8, 9, 18, 12, 3, 11, 10, 15, 5, 13, 10~
$ purpose        <chr> "credit_card", "debt_consolidation", "small_business", ~
$ home_ownership <chr> "OWN", "RENT", "RENT", "OWN", "RENT", "MORTGAGE", "MORT~
$ inq_last_6mths <int> 0, 0, 2, 1, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 3, 0~
```

```r
str(myLCdata)
```

```
Classes 'data.table' and 'data.frame':  44369 obs. of  5 variables:
 $ annual_inc    : num  60000 18576 40000 35000 125000 ...
 $ open_acc      : int  6 7 15 3 12 8 9 18 12 3 ...
 $ purpose       : chr  "credit_card" "debt_consolidation" "small_business" "home_improvemen
```

```
 $ home_ownership: chr  "OWN" "RENT" "RENT" "OWN" ...
 $ inq_last_6mths: int  0 0 2 1 0 2 0 0 1 0 ...
 - attr(*, ".internal.selfref")=<externalptr>
```

1.1 Variable Overview and Business Interpretation

The dataset `myLCdata` contains 44,369 observations and 5 variables. Based on the output of `glimpse()` and `str()`, the variables can be categorized as follows:

1.1.1 Variable Description

- **annual_inc (numeric)**
  The self-reported annual income provided by the borrower during registration. This variable is continuous and measures repayment capacity.
- **open_acc (integer)**
  The number of open credit lines in the borrower's credit file. This reflects the borrower's existing credit exposure.
- **purpose (character)**
  A category provided by the borrower for the loan request. (e.g., credit card refinancing, debt consolidation, small business, home improvement). This is a categorical variable.
- **home_ownership (character)**
  The home ownership status provided by the borrower during registration.(e.g., RENT, OWN, MORTGAGE, OTHER).
- **inq_last_6mths (integer)**
  The number of inquiries in past 6 months (excluding auto and mortgage inquiries). This is an indicator of recent credit-seeking behavior and potential financial stress.

1.2 Business Context in Peer-to-Peer Lending

<what to write here??????????>

2 Task 2: Data Understanding Based on Summary Statistics

2.1 Summary Statistics

```r
summary(myLCdata)
```

```
   annual_inc         open_acc       purpose           home_ownership
 Min.   :   4000   Min.   : 1.00   Length:44369       Length:44369
 1st Qu.:  45649   1st Qu.: 8.00   Class :character   Class :character
 Median :  65000   Median :11.00   Mode  :character   Mode  :character
 Mean   :  75269   Mean   :11.57
 3rd Qu.:  90000   3rd Qu.:14.00
 Max.   :6000000   Max.   :76.00
                   NA's   :2
 inq_last_6mths
 Min.   : 0.0000
 1st Qu.: 0.0000
 Median : 0.0000
 Mean   : 0.6966
 3rd Qu.: 1.0000
 Max.   :31.0000
 NA's   :2
```

```r
# install skimr if necessary
if (!require("skimr")) install.packages("skimr")
library(skimr)

skim(myLCdata)
```

Table 1: Data summary

Name	myLCdata
Number of rows	44369
Number of columns	5
Key	NULL
Column type frequency:	
character	2
numeric	3

Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
purpose	0	1	3	18	0	14	0
home_ownership	0	1	3	8	0	6	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
annual_inc	0	1	75269.01	60410.50	4000	45649	65000	90000	6000000	
open_acc	2	1	11.57	5.38	1	8	11	14	76	
inq_last_6mths	2	1	0.70	1.01	0	0	0	1	31	

2.2 Interpretation and Domain Context

2.2.1 1. Missing Values (NAs)

- Are missing values present?
- In which variables?
- How many?
- Is this plausible from a business perspective?
- Would this require data preparation?

Discussion: > Write interpretation here.

2.2.2 2. Numerical Variables

For each numeric/integer variable:

2.2.2.1 Variable: _____

- Mean:
- Median:
- Minimum:
- Maximum:
- Standard deviation:
- Is the distribution symmetric or skewed?
- Are there potential outliers?
- Do the extreme values seem realistic in the Lending Club context?

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.3 3. Categorical Variables

For each categorical variable:

2.2.3.1 Variable: _____

- Number of categories:
- Most frequent category:
- Least frequent category:
- Are categories balanced or highly unbalanced?
- Does this distribution make business sense?

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.4 4. Key Insights from Summary Statistics

- Most important statistical findings:
- Potential data quality issues:
- First hypotheses about borrower behavior:

---

3 Task 3: Univariate Exploratory Data Analysis (EDA) with ggplot2

3.1 Variable 1: _____

3.1.1 Visualization

```
# Insert ggplot code here
```

3.1.2 Why This Plot Type?

- Why is this diagram appropriate for the variable type?
- What can be read from this diagram type (e.g., range, distribution, frequency)?

3.1.3 Interpretation

- What does the distribution look like?
- Are there outliers?
- Is the distribution skewed?
- What does this imply in the business context?
- Does this confirm findings from Task 2?

3.2 Variable 2: _____

3.2.1 Visualization

```
# Insert ggplot code here
```

3.2.2 Interpretation

Write here.

3.3 Variable 3: _____

3.3.1 Visualization

```
# Insert ggplot code here
```

3.3.2 Interpretation

Write here.

3.4 Variable 4: _____

3.4.1 Visualization

```
# Insert ggplot code here
```

3.4.2 Interpretation

Write here.

3.5 Variable 5: _____

3.5.1 Visualization

```
# Insert ggplot code here
```

3.5.2 Interpretation

Write here.

3.6 Overall Observations from Univariate EDA

- Which variables show skewness?
- Which variables show imbalance?
- Any surprising patterns?
- Do visual findings confirm summary statistics?

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4 Task 4: Bivariate Exploratory Data Analysis (EDA)

4.1 Pairs Plot Overview

```r
if (!require("GGally")) install.packages("GGally")
library(GGally)

ggpairs(myLCdata)
```

```
Warning: Removed 2 rows containing missing values
Removed 2 rows containing missing values

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
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Warning: Removed 2 rows containing non-finite outside the scale range (`stat_bin()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
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`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

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Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).
```

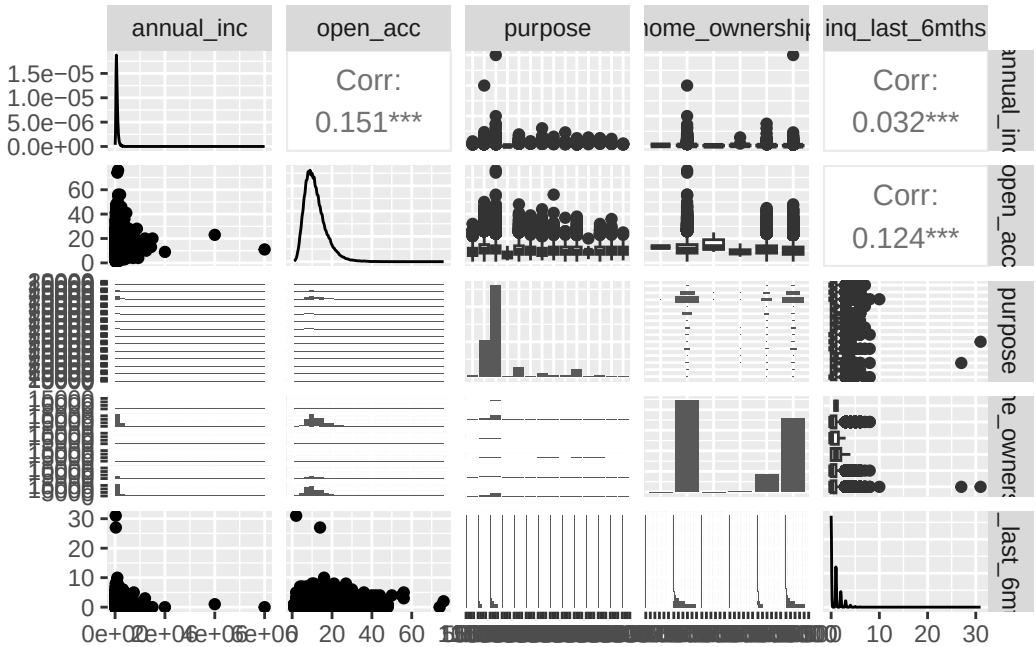

4.1.1 Initial Observations

- Are there visible correlations?
- Any clear patterns?
- Any surprising relationships?

4.2 Selected Insight 1

4.2.1 Variable Pair: _____ vs _____

4.2.2 Why This Pair Is Interesting

- What pattern was observed?
- Why was it not obvious before?

4.2.3 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.2.4 Interpretation

- What does this relationship indicate?
- Business interpretation?

4.3 Selected Insight 2

4.3.1 Variable Pair: _____ vs _____

4.3.2 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.3.3 Interpretation

Write here.

4.4 Adding a Third Variable

4.4.1 Enhanced Plot with Third Variable (e.g., color, shape, facet)

```
# ggplot with third aesthetic mapping
```

4.4.2 Why This Third Variable?

- Why is it relevant?
- What additional insight does it provide?
- What was not visible before?

4.5 Final Bivariate Insights

- Most important relationships discovered:
- Implications for credit risk assessment:
- Potential hypotheses for predictive modeling: